

Carbon Policy in Production Networks

Three Essays on Emissions, Networks, and Climate Policy

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Three essays on firms, emissions, and the production network

1. Inferring Firms' Emissions from Transactions Data

measurement: who emits, when emissions are unobserved?

(~10 min)

2. Carbon Leakage Within and Across Countries

reduced form: does carbon pricing push activity to dirtier firms?

(~10 min)

3. Carbon Policies in Production Networks

structural: how does *who you cover* shape aggregate emissions?

(~25 min)

A common thread: firm-level emissions and the buyer–supplier network,
measured in near-universal Belgian administrative data.

Chapter 1

Inferring Firms' Emissions from Transactions Data

with Gert Bijmans · ~10 minutes

Inferring emissions: the problem in one slide

Question: for the *universe* of firms — where emissions are unobserved — how far can **B2B transaction data** take us in inferring firm-year emissions?

Why hard: registries (EU ETS) & surveys cover only **large installations**, in a handful of countries

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Data: Belgium — near-universe B2B (2002–22) + **verified EU ETS emissions** as ground truth

Result: a cross-fitted **elastic-net proxy** separates emitters from non-emitters (TPR 96%, FPR $\leq 17\%$) and ranks them (Spearman 0.64)

Training sample: where the signal is learned

The set of firms we use to identify relevant suppliers:

- (i) **EU ETS firms**: 257 firms, 3,170 firm-years
 - verified yearly emissions from the EU Transaction Log
- (ii) **Non-ETS firms in ~100%-covered sectors**: 2,959 firms, 23,438 firm-years
 - Petroleum refining (19), Iron & steel (24), Paper, pulp & print (17/18)
 - assigned **zero combustion emissions** (EU ETS covers ~100%)

Why include the zeros? → gives the model **extensive-margin** information; same sectors are **energy-intensive** ⇒ a demanding test.

Let the data identify the relevant suppliers

- ▶ **Benchmark**: sum purchases from the six fuel-selling NACE codes (Tabachova 2025, Stangl et al. 2026) — misses multi-activity sellers, symmetric weights
- ▶ **This paper**: regress verified emissions on **firm-year purchases from each supplier**

$$y_{it} = \sum_j \beta_j x_{ijt} + \gamma \log(\text{revenue}_{it}) + \delta' \mathbf{d}_t + \phi' \mathbf{s}_i + \varepsilon_{it}$$

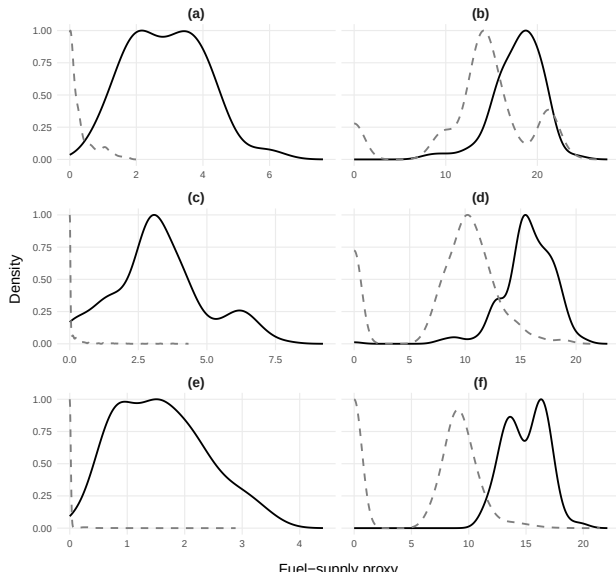
$y_{it} = \text{asinh}(\text{emissions})$, $x_{ijt} = \text{asinh}(\text{sales})$; revenue & FEs **unpenalized**

- ▶ **Elastic-net penalty** on $\{\beta_j\}$ ($\alpha = 0.5$, λ by firm-grouped CV) $\Rightarrow$
EN-proxy $_{it} = \sum_{j: |\hat{\beta}_j| > 0} \hat{\beta}_j x_{ijt}$

Who are the selected suppliers?

- ▶ Out of $\sim 22,700$ candidates, EN selects **612 suppliers** (463 +, 149 –)
 - ▶ Dispersed across **53 NACE 2-digit sectors**:
 - largest: **wholesale trade (46)**; then construction, engineering, transport
 - **notably absent**: refined petroleum (19), electricity/gas (35)
 - ▶ Overlap with the NACE fuel-seller set is **minimal**: only **4 of 612** (0.7%)
- ⇒ **data-driven selection and NACE codes pick nearly disjoint sets of firms**

The proxy separates emitters from non-emitters



Kernel densities of the proxy

Black: EU ETS firms

Grey: non-ETS firms

AUC (EN-proxy):

- Petrol. ref.: **0.99**
- Iron & steel: **0.97**
- Paper/print: **0.99**

Out-of-sample performance (sector-level cross-fitting)

	Prediction error			Correlation		Extensive margin			
	RMSE	nRMSE	MAPD	Levels	Rank	FPR	TPR	p50	p99
Revenue	190	1.00	0.71	0.85	0.53	1.00	1.00	0.01	0.39
Elastic Net	218	1.15	0.84	0.85	0.64	0.17	0.96	0.01	0.19
NACE	214	1.13	0.85	0.81	0.52	0.67	0.99	0.01	0.46
Gated Rev.	193	1.01	0.71	0.84	0.68	0.17	0.96	0.00	0.35
Geom. Mean	218	1.15	0.74	0.83	0.63	0.26	0.97	0.01	0.28

- ▶ EN dominates the **extensive margin** (FPR 0.17 vs. revenue 1.00, NACE 0.67)
- ▶ Revenue dominates **level accuracy among emitters** (MAPD 0.71)
- ▶ **Gated Revenue** (EN classifies, revenue scales) = best of both; rank corr. 0.68

Means over $M = 200$ sector-level CV repeats.

Chapter 1 — takeaways

- ▶ **B2B transaction data carry a signal about emissions** unavailable from balance-sheet variables alone
 - ▶ A **cross-fitted elastic-net proxy** beats NACE classification, and a hybrid recovers the best of both margins
 - ▶ **Broader lesson:** combustion $\Rightarrow$ fuel purchases $\Rightarrow$ network traces
 - a **general economic logic**, not Belgium-specific
 - testable wherever **transaction data + a partial registry** coexist
- $\hookrightarrow$ the same firm-level emissions & network reappear, structurally, in Chapter 3.

Chapter 2

Carbon Leakage Within and Across Countries

with Gert Bijmans · ~10 minutes

Carbon leakage: the problem in one slide

Question: does the EU ETS push activity toward *less-regulated* suppliers — **within** Belgium (to less carbon-exposed domestic firms) and **across** its borders (to non-EU imports)?

Data: firm-level Belgium — near-universe **B2B network** + firm $\times$ product $\times$ country
customs + **EUTL** emissions/allowances

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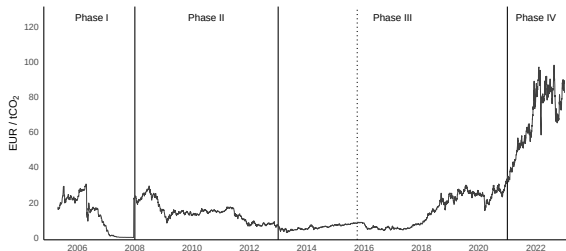
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Why: the relative-price gap stays **below 1%** even at full pass-through — too small to repay switching 11/33

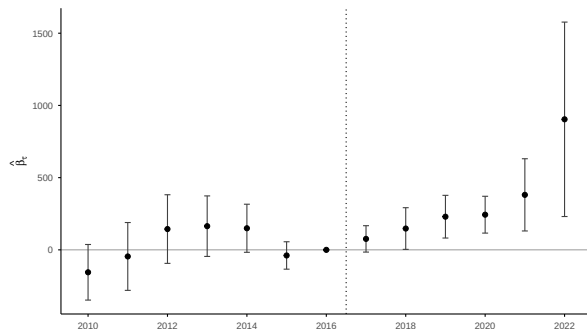
Setting: a common carbon price and an exogenous shock



- ▶ **EU ETS**: common European carbon price since 2005
- ▶ **Market Stability Reserve** (Oct 2015): rules-based, withdraws surplus allowances
 - mechanical, **predetermined**
 - not Belgian conditions
- ▶ Drives EUA **€5–8 (2015) → €25 (2018) → €80+ (2022)**
 - ⇒ the post-2017 run-up is the quasi-experiment

$$\text{Treatment dose } \omega_i = \frac{\max(0, \text{emissions}_i - \text{free allowances}_i) \cdot \text{EUA}}{\text{EUA}}, \text{ fixed at 2015–16.}$$

First: the policy does bite on prices

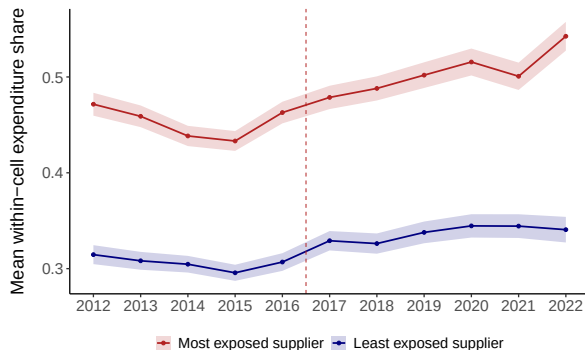


- ▶ Manufacturing PPI rises after a carbon shock and **stays elevated**
- ▶ Concentrated in **shortage-exposed** sectors (allowances bought on the market), **not** gross emissions

Cross-sector DiD, per SD of exposure:

- shortage ω^{sh} : **+0.012**** (to 2019)
- $\rightarrow$ **+0.023***** (to 2022)
- gross emissions: **null**

Margin 1 — within Belgium: no reallocation



- ▶ Within (buyer, NACE-4d): expenditure share on **most**- vs. **least**-exposed supplier

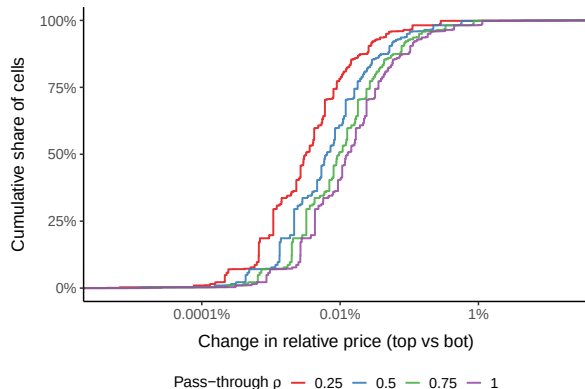
- ▶ Shares move in **parallel** after 2017

DiD on $\text{post} \times \text{top}$:

- naive: **+0.0123**** (wrong sign!)
- + $\text{age} \times \text{top}$ FE: **0.0031**, n.s.

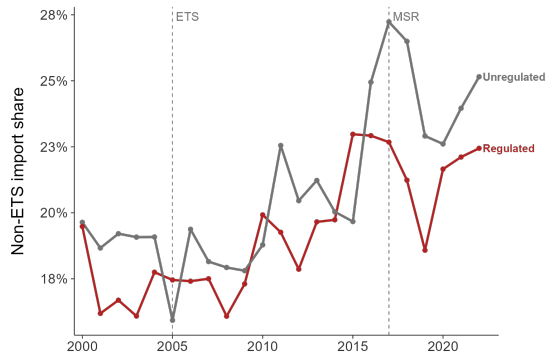
÷4 once mechanical attrition (top suppliers survive longer) is absorbed;
not robust under HonestDiD from

Why nothing to find: the incentive is tiny



- ▶ Map the ΔEUA into a relative-price gap between most- and least-exposed supplier
- ▶ Even at **full pass-through**, the median cell's gap is **well below 1%**
 - ⇒ substitution elasticity σ is unidentified — the price barely moves, so there is nothing to substitute against

Margin 2 — across borders: Belgium $\neq$ France



- ▶ Replicate Coster-Méjean-di Giovanni on Belgian customs: regulated imports, EU vs. non-EU sources
- ▶ Regulated $\times$ non-ETS share is **flat** across 2000–2022

Intensive margin (value share):

- Belgium: **+0.0004** (n.s.)
- France: **+0.121** ($p < 0.001$)

$\Rightarrow$ a common EU carbon price need **not** produce the same leakage everywhere

Chapter 2 — three margins, three nulls

The set of margins with detectable leakage is **empty**.

1. Pass-through is **real** and persistent — more in shortage-exposed sectors
2. **No** within-NACE reallocation: wrong-signed, $\div 4$ and insignificant, not robust
3. **No** international reallocation: flat shares; Belgium two orders of magnitude below France

Mechanism: the cost shock is small and pass-through partial $\Rightarrow$ the relative-price gap never clears the cost of switching suppliers.

$\hookrightarrow$ Chapters 1–2 measure firms and test reallocation *reduced-form*; Chapter 3 builds the structural model that says *when* reallocation should bind.

Chapter 3

Carbon Policies in Production Networks

Job Market Paper, with Gert Bijmans · ~25 minutes

Carbon policy in networks: the problem in one slide

Question: a carbon price covers *some* firms, not all. What is its effect on aggregate emissions, how does that depend on *whom* it covers, and is there a firm whose coverage delivers the largest reduction?

Theory: GE production network with heterogeneous emission intensities + costly abatement; a carbon price p_z and a 0/1 coverage vector τ

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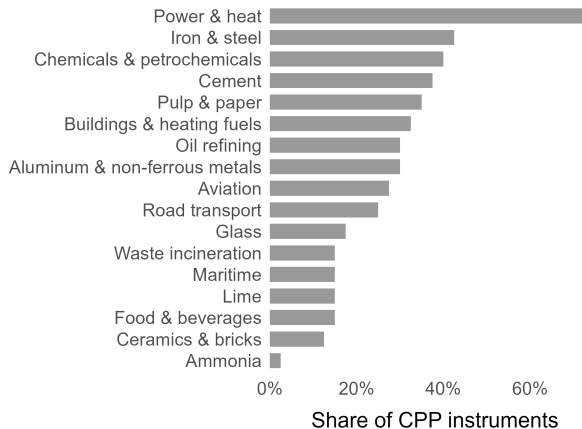
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unchline: $\sim 94\%$ of the gain from *whom to cover* comes from firms' **own** emissions — a

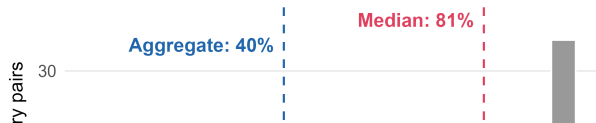
Carbon pricing is incomplete by design



► ~75 carbon-pricing instruments in 2024, ~a **quarter** of global emissions

► Coverage is **partial and uneven** across *and* within sectors

► EU ETS: median targeted sector ~80% priced, but only ^{20/33} ~**40%** of emissions priced in



Why the production network matters

Three questions: (1) effect of a scheme on aggregate emissions? (2) how does it depend on *who* is covered? (3) is there a firm whose coverage reduces emissions most?

- ▶ A firm's emissions are partly **its own**, partly **embodied in the inputs it buys**

1. **Who is exposed** to a carbon price is a network object (taxing one firm reaches beyond it)

2. The **aggregate effect** depends on where dirty firms sit in the network

The object that matters: **network-adjusted emission intensity**

$$\Psi_e := \Psi \bar{\mathcal{E}}, \quad \Psi = (I - \Omega)^{-1} \quad (\text{direct} + \text{upstream-embodied}; \text{scope } 1+2+3)$$

Environment

- ▶ Closed economy: household, government, N producers (CRS, marginal-cost pricing $p_i = mc_i$)
- ▶ **Policy**: carbon price p_z and coverage vector $\tau = \{\tau_i\}$, $\tau_i \in \{0, 1\}$
- ▶ Producer i : Cobb–Douglas in labor and a **nested-CES** input bundle
 - across sectors σ_B (weak substitution), within sector σ_W (stronger)
- ▶ **Network objects**: input–output matrix Ω , Domar weights λ_i , Leontief inverse $\Psi = (I - \Omega)^{-1}$
- ▶ Emissions a by-product of production: $z_i = e_i f_i$, intensity $\bar{e}_i$ in the no-policy economy

Firms abate: the technique channel

- ▶ A covered firm pays p_z per unit of emissions $\Rightarrow$ it **abates**: chooses effort κ_i

$$e_i = (1 - \kappa_i)^{\alpha_i} \bar{e}_i$$

α_i governs **how hard firm i is to abate** (disciplined from abatement-elasticity estimates)

- ▶ To first order, the **technique response**:

$$\left. \frac{d \log e_i}{dp_z} \right|_{p_z=0} = -\tau_i \alpha_i^2 \bar{e}_i < 0$$

- ▶ **Key separability**: abatement lowers intensity but leaves the firm's *input mix* (hence pass-through) unchanged to first order

The carbon price propagates through prices

- ▶ Each firm passes its carbon cost into its price; buyers inherit it through their inputs:

$$\frac{d \log p_i}{dp_z} = \sum_{j \in \mathcal{N}} \Omega_{ij} \frac{d \log p_j}{dp_z} + \Omega_{iL} \frac{d \log w}{dp_z} + \tau_i \bar{e}_i$$

- ▶ In matrix form (the **exposure** of every price to the policy):

$$\frac{d \log \mathbf{p}}{dp_z} = \Psi (\tau \circ \bar{\mathcal{E}}) + \Psi_{(:,L)} \frac{d \log \lambda_L}{dp_z}$$

⇒ a firm's exposure is its *covered, network-weighted* upstream emission content
— itself a Ψ object

Decomposition: scale, technique, reallocation

- ▶ Aggregate emissions move through three channels (Grossman–Krueger accounting, mapped to primitives):

$$\left. \frac{d \log Z}{dp_z} \right|_0 = \underbrace{\sum_{i \in \mathcal{N}} \frac{z_i}{Z} \frac{d \log e_i}{dp_z}}_{\text{technique (abatement)}} + \underbrace{\sum_{i \in \mathcal{N}} \frac{z_i}{Z} \left[\frac{d \log \lambda_i}{dp_z} - \frac{d \log p_i}{dp_z} \right]}_{\text{reallocation}}$$

- ▶ **Scale** ≈ 0 : by Hulten, $\left. \frac{d \log Y}{dp_z} \right|_0 = 0$ — to first order the policy is a pure pollution-cutting device, not an output story
- ▶ **Technique** is set by *covered* firms; **reallocation** is where the network enters

Main result: a sufficient-statistic decomposition

$$\left. \frac{d \log Z}{dp_z} \right|_{p_z=0} = \underbrace{- \sum_{i \in \mathcal{N}} \frac{\tau_i z_i}{Z} \alpha_i^2 \bar{e}_i}_{\text{technique} < 0, \text{ covered firms}} - \sigma \underbrace{\sum_{i \in \mathcal{N} \cup \{C\}} \frac{x_i}{Z} \mathbb{C}_{\Omega(i,:)} \left(\frac{d \log \mathbf{p}}{dp_z}, \Psi_e \right)}_{\text{reallocation: one covariance}}$$

- ▶ Across each buyer's inputs: covariance between **exposure** ($\frac{d \log \mathbf{p}}{dp_z}$) and **network-adjusted intensity** Ψ_e
 - ▶ Its **sign** decides whether the network **amplifies** ($\mathbb{C} > 0$) or **leaks** ($\mathbb{C} < 0$)
 - ▶ $(1 - \sigma)$ governs the **magnitude**, not the sign
- Complete coverage** ($\tau = 1$): covariance $\rightarrow$ **variance** $\Rightarrow$ reallocation *always* amplifies

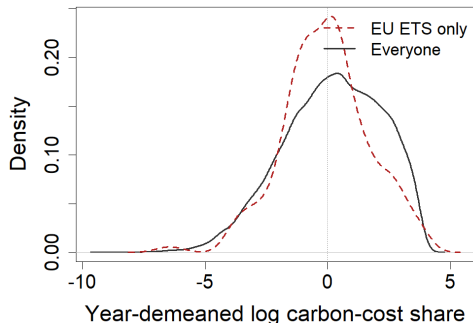
Whom to cover? Emission centrality

- ▶ The decomposition is **additively separable** across covered firms $\Rightarrow$ define the marginal value of covering firm j alone:

$$\text{Centrality}_j = -\frac{z_j}{Z} \alpha_j^2 \bar{e}_j - \sigma \sum_{i \in \mathcal{N} \cup \{C\}} \frac{x_i}{Z} \mathbb{C}_{\Omega_{(i,:)}} \left(\Psi_{(:,j)} \bar{e}_j, \Psi_e \right)$$

- ▶ **Own-abatement** term $(z_j, \bar{e}_j)$ + a **network-position** term
- ▶ This is the object counterfactuals **rank firms on** — and it makes the network's value an *empirical* question

Taking it to Belgian data: facts that discipline the model



- ▶ At $p_z = 80$: median emitter's carbon cost $\approx$ **2.5%** of total cost; ~ 1 in 6 exceed 20%
- ▶ Wide **dispersion** in direct intensity: within (4d, year) $p_{90}/p_{10} \approx 5.7$; $\rightarrow \sim 30$ -fold with the imputed tail
- ▶ Direct $\leftrightarrow$ upstream link is **weak** (Spearman 0.24–0.33)
- ▶ For **1 in 3** ETS firm-years, *upstream* emissions exceed own direct

Thin within-sector supplier base (~ 1.5 suppliers/sector) vs. broad across-sector base (~ 25 sectors sourced) $\Rightarrow \sigma_W > \sigma_B$.

Calibration

- ▶ 2019 Belgian firm-level network (Ω, λ) from B2B + imports + balance sheets; intensities $\bar{\mathcal{E}}$ from EUTL (+ imputation for the non-ETS tail)
- ▶ Realized **EU ETS** coverage τ ; carbon price $p_z = \mathbf{80\ EUR/t}$ (2022 EUA)
- ▶ Benchmark elasticities:

$$\sigma_B = 0.1, \quad \sigma_W = 2.5, \quad \rho = 0.7, \quad \alpha = 2$$

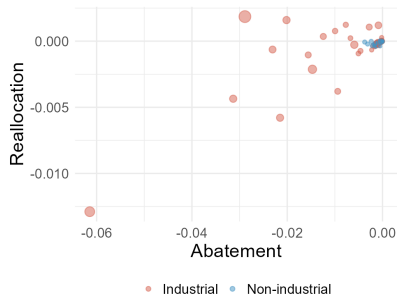
(ρ : incomplete pass-through, a Katz–Bonacich Leontief Ψ_ρ ; α from abatement-elasticity estimates. Robustness over grids on each.)

Result 1 — the EU ETS effect is mostly abatement

Channel	$\Delta \log Z$	share
Total	-0.421	
Scale	-0.000	≈ 0
Technique	-0.371	$\approx 88\%$ ($\sim 7/8$)
Composition	-0.049	$\approx 12\%$

- ▶ Composition is **negative** $\Rightarrow$ the Belgian network **amplifies** abatement (covariance positive: covered firms are positioned in cleaner-leaning supply chains)
- ▶ But **uneven**: taxing a firm lowers emissions at **~ 4 in 5** firms, raises them at the rest — the raisers offset **$\sim 1/5$** of the reductions
- ▶ For **~ 1 in 25** firms, coverage *backfires*: demand shifts to dirtier uncovered rivals

Result 2 — which firms beats how many



- ▶ Re-selecting the *same number* of firms toward central ones: **+31.7%** vs. ETS
- ▶ Expanding to *all* of industry: only **+5.3%**

⇒ **which $\approx 6\times$ how many**

Technique vs. composition centrality (corr. 0.74).

But: holding firm count fixed, ranking on **direct emissions z_i alone** recovers **$\sim 94\%$** of the centrality gain — the network's extra value sits at the *backfiring* margin, not in everyday selection.

Chapter 3 — takeaways

- ▶ A **transparent decomposition** of targeted carbon pricing in a network: scale (≈ 0), technique, and reallocation — the last a **single covariance** of exposure with network-adjusted intensity Ψ_e
 - ▶ Empirically (Belgium, EU ETS): the cut is **$\sim 7/8$ abatement**; the network **amplifies** modestly
 - ▶ The network is indispensable for **diagnosis** — measuring the effect and flagging the ~ 1 -in-25 firms that backfire — but for **selection** a constrained regulator simply **targets the heaviest emitters**
- Next:** open economy (cross-border leakage — Chapter 2), and the jointly-optimal covered set

Thank you!

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